Charles Boalch

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EDUCATION

University of Georgia, Morehead Honors College, Athens, GA — May 2029
B.S. Computer Science, B.S. Statistics — GPA: 3.96 / 4.0
Relevant Coursework: Statistical Methods, Data Structures, Systems Programming

Georgia Institute of Technology, Atlanta, GA — August 2024 – May 2025
Dual Enrollment Student — GPA: 3.5 / 4.0
Relevant Coursework: Introduction to Object Oriented Programming

Paul Duke STEM High School, Norcross, GA — August 2021 – May 2025
2025 Gifted Student of the Year, Norcross Swim and Dive Captain — GPA: 4.0 / 4.0

WORK EXPERIENCE

Peachtree Station Swim & Tennis — Peachtree Corners, GA
Assistant Swim Coach — May 2026 – July 2026

- Worked with elementary through high school children to build essential stroke skills
- Assisted head coach with routine practice duties and swim meet organization
- Handled emergencies with swimmers promptly and smoothly as to not disrupt swim meets

Pacific Diversified Services — Norcross, GA
Machine Operator — April 2024 – August 2025

- Operated and monitored specialized machinery for fiber optic spool cleaning
- Developed custom Python applications to automate and streamline repetitive managerial tasks
- Facilitated communication and workplace engagement between coworkers of different languages

PROJECT EXPERIENCE

Swimming Finals Predictor — April 2026

- Wrote a Python script to collect over 3,500 data points from a national-level swim meet
- Used R to create a multivariate logistic regression model predicting if a swimmer would qualify for a championship final
- Evaluated using a test data set to show 93.5% model accuracy and high predictive reliability

Song to Movie — December 2025

- Collected large amounts of movie data and scored using AI models across six emotional dimensions
- Matched analyzed songs to similar movies by using cosine and L2 similarity metrics in movie space
- Deployed a React web interface that integrates Spotify API to give a personalized user experience

UGA Dining Analysis — September 2025 – November 2025

- Scraped dining hall capacity information from a UGA endpoint over a multiweek period
- Analyzed time series information to make optimized decisions on when to eat meals

CAMPUS & COMMUNITY INVOVEMENT

Club Swim Team, University of Georgia — Athens, GA
Athlete — August 2025 – present

- Write rigorous, research-based exercise plans to build swimmer strength and endurance
- Develop a React website to accurately predict future swimmer times using a Bayesian model

Coaching4Success, UGA Office of Service-Learning — Athens, GA
Student Volunteer — September 2025 – December 2025

- Oversaw and participated in engaging and educational lessons with elementary-aged students
- Helped with post-event maintenance operations and coordinate with school administration